Course Syllabus: CS-401 Machine Learning

1. Course Details

Course Code: CS-401
Course Title: Introduction to Machine Learning and Data Science
Credits: 4 (L: 3, T: 1, P: 0)
Prerequisites: Linear Algebra, Probability & Statistics, Python Programming

2. Course Objective

This course introduces core concepts of machine learning, including supervised and unsupervised learning algorithms, neural networks, model evaluation metrics, and practical applications in python using scikit-learn and tensorflow.

3. Syllabus Units

Unit 1: Introduction to Data Preprocessing, Feature Engineering, and Exploratory Data Analysis (EDA) (Weeks 1-3)
Unit 2: Supervised Learning - Linear and Logistic Regression, Decision Trees, Random Forests, Support Vector Machines (Weeks 4-7)
Unit 3: Unsupervised Learning - K-Means Clustering, Hierarchical Clustering, Principal Component Analysis (PCA) (Weeks 8-10)
Unit 4: Introduction to Neural Networks, Backpropagation, and Deep Learning Basics (Weeks 11-13)
Unit 5: Model Deployment, Ethics in AI, and Final Project Presentations (Week 14)

4. Grade Breakdown

- Assignments & Quizzes: 20%
- Mid-Semester Examination: 30%
- Term Project & Lab Assessments: 20%
- End-Semester Final Examination: 30%